

Physics Department *Fisika Departement*

Semester Test 2

Semestertoets 2

FSK116

17 May 2014 / 17 Mei 2014

Surname and initials	MEMO	Student number	MEMO
Van en voorletters		Studentenommer	
Signature	MEMO		
Handtekening			
Time	2½ hours	Total	65 marks
Tyd	2½ ure	Totaal	65 punte
Internal Examiner:	Dr J M Nel	External Examiner	Mr R Q Odendaal
Interne Eksaminator		Eksterne Eksaminator	

- Answer all questions in the space provided in PEN. - Clearly show all calculations. Do not just write down the final answer. - Clearly state all assumptions, rules and laws used. - Where applicable draw the free-body diagrams. - Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks. - Programmable calculators may NOT be used. NO textbooks allowed. - A cheat sheet may be used, but must adhere to the rules printed on it. - All test and examination regulations of the University of Pretoria are applicable.	<i>Beantwoord alle vrae in die spasie gelaat in PEN.</i> <i>Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.</i> <i>Stel duidelik alle aannames, reëls en wette wat u gebruik.</i> <i>Waar van toepassing, teken die vryliggaamsdiagram.</i> <i>Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.</i> <i>Programmeerbare sakrekenaars MAG NIE gebruik word nie.</i> <i>GEEN handboeke word toegelaat nie.</i> <i>'n Formuleblad mag gebruik word onderhewig aan die reëls daarop gedruk.</i> <i>Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.</i>
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Signature		Venue	
Handtekening		Lokaal	

MEMO

Question 1/Vraag 1

A fly collides with the windscreen of a moving bus.

'n Vlieg tref die voorruit van 'n bewegende bus.

- a) Which of the two experiences a greater force due to the collision, the fly or the bus? Explain.

Watter van die twee ondervind die groter krag as gevolg van die botsing, die vlieg of die bus?

Verduidelik.

[3]

They both experience the same force. ✓ This comes from Newton's third law
If a body pushes on a second body, then the second body pushes back on the
First with the force equal in magnitude, but opposite in direction ✓

- b) Which experiences the greater acceleration?

Watter van die twee ondervind die groter versnelling? Verduidelik.

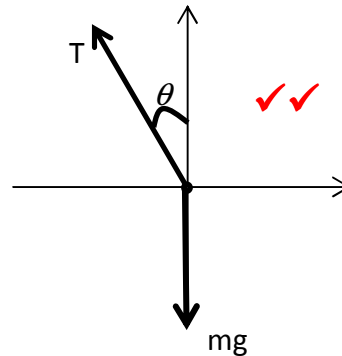
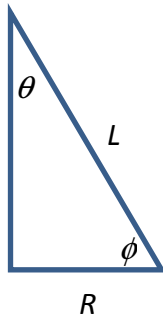
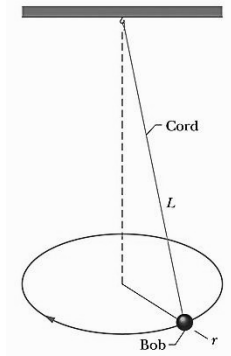
[3]

The fly will experience the greater acceleration. ✓
From the relationship $F = ma$, with the forces equal, the small mass ✓ of the
Fly in comparison to the bus, makes it necessary for the acceleration of the
fly to be greater than that of the bus. ✓

Question 2/Vraag 2

The figure shows a *conical pendulum*, in which the bob (the small object at the lower end of the cord) moves in a horizontal circle at constant speed. (The cord sweeps out a cone as the bob rotates.) The bob has a mass of 0.040 kg, the string has length $L = 0.90$ m and has negligible mass, and the bob follows a circular path of circumference 0.94 m.

Die figuur dui 'n koniese slinger aan, waar die klein massa onderaan die tou in 'n horisontale sirkel beweeg teen 'n konstante spoed. (Die tou teken 'n kegelvorm in die ruimte soos die massa roteer). Die massa is 0.040 kg, en die tou het 'n lengte $L = 0.90$ m met 'n weglaatbare massa. Die massa beweeg in 'n sirkelbaan met 'n omtrek van 0.94 m.



- a) Determine the tension in the string.
Bepaal die spankrag in die tou.

[2+4]

The radius of the circle which the bob travels is R . The circumference of the circular path is given and therefore:

$$\text{circumference} = 2\pi R$$

$$R = \frac{0.94 \text{ m}}{2\pi} = 0.15 \text{ m} \quad \checkmark$$

Determine either the angle the cord makes with the horizontal or the vertical axis. The angle that the cord makes with the horizontal is now easily found:

$$\phi = \cos^{-1}(R/L) = \cos^{-1}(0.15 \text{ m}/0.90 \text{ m}) = 80.4^\circ.$$

The angle that the cord makes with the vertical is now easily found:

$$\theta = \sin^{-1}(R/L) = \sin^{-1}(0.15 \text{ m}/0.90 \text{ m}) = 9.6^\circ.$$

} $\checkmark$ one of these angles

Note that here we are using T for the tension in the cord.

$$\sum F_y = T \cos \theta - mg = ma_y = 0 \quad \checkmark$$

$$\begin{aligned} \therefore T &= \frac{mg}{\cos \theta} \\ &= \frac{(0.4 \text{ kg})(9.8 \text{ m/s}^2)}{\cos 10^\circ} \\ &= 0.40 \text{ N} \quad \checkmark \end{aligned}$$

- b) Determine the period of the motion.
Bepaal die periode van die beweging.

[5]

The horizontal component of that tension must supply the net acceleration for the circular motion,

$$\sum F_x = T \sin \theta = ma_x = m \frac{v^2}{R} \quad \checkmark$$

$$v = \sqrt{\frac{TR \sin \theta}{m}} = \sqrt{\frac{gR \sin \theta}{\cos \theta}} \quad \checkmark$$

$$v = 0.51 \text{ m/s} \quad \checkmark$$

At this speed, the time to complete one revolution is:

$$v = \frac{\text{distance}}{\text{time}} \quad \checkmark$$

$$\text{time} = \frac{\text{distance}}{v} = \frac{2\pi R}{v} \quad \checkmark$$

This divided into the circumference gives the time for one

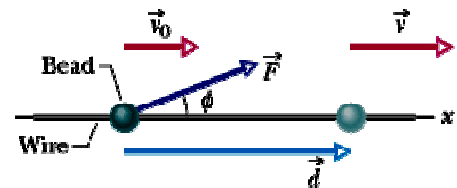
$$= \frac{2\pi(0.15 \text{ m})}{0.51 \text{ m/s}} \quad \checkmark$$

$$= 1.8 \text{ s}$$

$$\text{revolution: } 0.94/0.51 = 1.8 \text{ s}$$

Question 3 / Vraag 3

Consider a bead, with a mass m that can slide along a frictionless wire that is stretched along a horizontal x axis. A constant force $\vec{F}$, directed at an angle ϕ with respect to the wire, accelerates the bead along the wire. Using Newton's second law and the associated acceleration, clearly show the relationship that exists between the work done by the force on the particle and the change in the kinetic energy of the particle. Explain each step and state any assumptions you may make.



Beskou 'n kraletjie, met massa m , wat op 'n wrywinglose draadjie, wat langs die horisontale x -as gespan is, gly. 'n Konstante krag, $\vec{F}$, wat teen 'n hoek ϕ met betrekking tot die draadjie gerig is, versnel die kraletjie langs die draadjie. Gebruik Newton se tweede wet, en die geassosieerde versnelling, om die verband tussen die arbeid verrig deur die krag op die deeltjie en die verandering in die kinetiese energie van die deeltjie te kry. Verduidelik elke stap en enige aannames wat jy maak.

[5]

This comes directly out of my lecture notes on clickUP or from the Textbook in Ch 7.

Relating the force and the acceleration experienced by the bead, from newton's 2nd law:

$$F_x = ma_x \quad \textcircled{1} \quad \checkmark$$

$$v^2 = v_o^2 + 2a_x d \quad \checkmark$$

$$a = (v^2 - v_o^2) / 2d$$

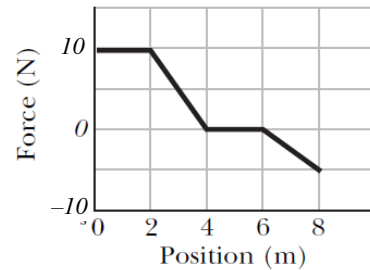
Substitute above in $\textcircled{1}$

$F_x = m ((v^2 - v_o^2)/2d)$
$F_x d = \frac{1}{2} m (v^2 - v_o^2)$ ✓
Then $F_x = \frac{1}{2} m (v^2 - v_o^2) = K_f - K_i$ ✓
But $F_x = F \cos \phi = F \cdot d = W$
Then $W = K_f - K_i = \Delta K$ ✓

Question 4 Vraag 4

A 2.5 kg block initially at rest, moves in a straight line on a horizontal frictionless surface under the influence of a force that varies with position of the block, as shown in the figure. Determine the speed of the block as it moves through the 8.0 m mark.

'n 2.5 kg blok, aanvanklik in rus, beweeg in 'n reguit lyn oor 'n horisontale wrywinglose oppervlak onder die invloed van 'n krag wat varieer met die blok se posisie, soos aangedui in die figuur. Bepaal die spoed van die blok wanneer dit verby die 8.0 m merk beweeg.



[6]

The speed can be determined using the kinetic energy work theorem: $W = K_f - K_i = \Delta K$

The net work can be determined from the area under the F vs distance curve which is given.

Work 0 – 2 m: $W_{0-2} = Fd = 10 \text{ N} \times 2 \text{ m} = 20 \text{ J}$ ✓

Work 2 – 4 m: $W_{2-4} = \vec{F} \cdot \vec{d} = \frac{1}{2}(10 \text{ N} \times 2 \text{ m}) = 10 \text{ J}$ ✓

Work 4 – 6 m: $W_{4-6} = \vec{F} \cdot \vec{d} = (10 \text{ N} \times 0 \text{ m}) = 0 \text{ J}$ ✓

Work 6 – 8 m: $W_{6-8} = \vec{F} \cdot \vec{d} = \frac{1}{2}(-5 \text{ N} \times 2 \text{ m}) = -5 \text{ J}$ ✓

Net work: $W_{\text{net}} = 20 \text{ J} + 10 \text{ J} + 0 \text{ J} - 5 \text{ J} = 25 \text{ J}$

Work kinetic energy theorem

$$W_{\text{net}} = \Delta K = K_f - K_i$$

$$K_f - K_i = 25 \text{ J} \quad \checkmark$$

$$K_f - 0 = \frac{1}{2}mv^2 = 25 \text{ J}$$

$$v = \sqrt{\frac{25}{\frac{1}{2}m}} = 20 \text{ m/s} \quad \checkmark$$

Question 5 / Vraag 5

A 5.0 g marble is fired vertically upward using a spring gun. The spring must be compressed by 8.0 cm if the marble is to just reach a target 20 m above the marble's position **on the compressed** spring.

'n 5.0 g albaster word vertikaal opwaarts gevuur uit 'n veer-geweer. Die veer moet 8.0 cm saamgepers word sodat die albaster net-net die teiken bereik wat 20 m bo die albaster se posisie **op die saamgepersde** veer is.

- a) Determine the change in the gravitational potential energy of the marble-Earth system during the 20 m ascent.

Bepaal die verandering in die swaartekrag-potensiële energie van die albaster-Aarde stelsel tydens die 20 m opwaartse beweging.

[3]

We take the reference point for gravitational potential energy at the position of the marble when the spring is compressed.

$U = 0\text{ J}$ at this point. ✓

The gravitational potential energy when the marble is at the top of its motion is $U_g = mgh$, ✓

where $h = 20\text{ m}$ is the height of the highest point. Thus,

$$U_g = (5.0 \times 10^{-3}\text{ kg})(9.8\text{ m/s}^2)(20\text{ m}) = 0.98\text{ J}.$$

Change in gravitational potential energy is 0.98 J ✓

-
- b) Determine the spring constant of the spring.

Bepaal die veerkonstante van die veer.

[5]

$$\Delta K + \Delta U_g + \Delta U_s = 0, \quad \checkmark$$

The kinetic energy is zero at the release point and at the highest point therefore $\Delta K = 0\text{ J}$. ✓

ΔU_s is the change in the spring's elastic potential energy, and ΔU_g is the change in the gravitational potential energy, then conservation of mechanical energy implies

$$\Delta U_s = -\Delta U_g = -0.98\text{ J}. \quad \checkmark$$

We take the spring potential energy to be zero when the spring is relaxed.

Then, our result in the previous part implies that its initial potential energy is $U_s = 0.98\text{ J}$. Then

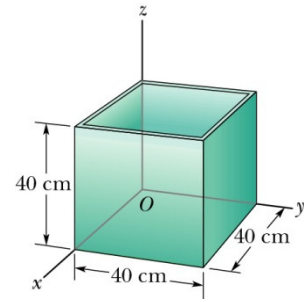
$U_s = \frac{1}{2}kx^2$, where k is the spring constant and x is the initial compression. Consequently,

$$k = \frac{2U_s}{x^2} = \frac{2(0.98\text{ J})}{(0.080\text{ m})^2} = 3.1 \times 10^2\text{ N/m} = 3.1\text{ N/cm}. \quad \checkmark$$

Question 6 / Vraag 6

A cubical box is constructed from uniform metal plate with negligible thickness. The box is open at the top and has an edge length $L = 40$ cm. Find the coordinates of the centre of mass of the box.

'n Kubieke boks word van uniforme metaal plaat met 'n weglaatbare dikte gemaak. Die boks is oop aan die bokant, en het 'n sylengte van $L = 40$ cm. Bereken die koördinate van die massamiddelpunt van die boks.



[9]

The centers of mass (with centimeters understood) for each of the five sides are as follows:

$$(x_1, y_1, z_1) = (0, 20, 20) \quad \text{for the side in the } yz \text{ plane}$$

$$(x_2, y_2, z_2) = (20, 0, 20) \quad \text{for the side in the } xz \text{ plane}$$

$$(x_3, y_3, z_3) = (20, 20, 0) \quad \text{for the side in the } xy \text{ plane}$$

$$(x_4, y_4, z_4) = (40, 20, 20) \quad \text{for the remaining side parallel to side 1}$$

$$(x_5, y_5, z_5) = (20, 40, 20) \quad \text{for the remaining side parallel to side 2}$$

Recognizing that all sides have the same mass m , we have the following.

The x coordinate of the center of mass is

$$x_{\text{com}} = \frac{mx_1 + mx_2 + mx_3 + mx_4 + mx_5}{5m} = \frac{0 + 20 + 20 + 40 + 20}{5} = 20 \text{ cm} \quad \checkmark \checkmark \checkmark$$

The y coordinate of the center of mass is

$$y_{\text{com}} = \frac{my_1 + my_2 + my_3 + my_4 + my_5}{5m} = \frac{20 + 0 + 20 + 20 + 40}{5} = 20 \text{ cm}$$

$\checkmark \checkmark \checkmark$

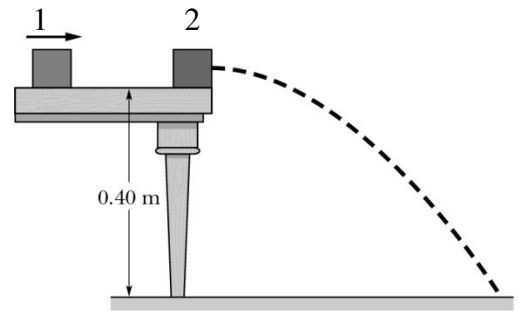
The z coordinate of the center of mass is

$$z_{\text{com}} = \frac{mz_1 + mz_2 + mz_3 + mz_4 + mz_5}{5m} = \frac{20 + 20 + 0 + 20 + 20}{5} = 16 \text{ cm} \quad \checkmark \checkmark \checkmark$$

Question 7 / Vraag 7

A 3.2 kg box (1) slides on a horizontal frictionless table and collides with a second box (2) of 2.0 kg, initially at rest on the edge of the table, at a height $h = 0.40$ m. The two boxes collide and stick together due to sticky tape on their sides and together they hit the floor with a kinetic energy of 29 J. Determine the initial speed of the 3.2 kg box.

'n 3.2 kg boks (1) gly oor 'n horisontale wrywinglose tafel en bots met 'n tweede boks (2) van 2.0 kg, wat aanvanklik in rus is op die rant van die tafel, op 'n hoogte $h = 0.40$ m. Die twee bokse bots en klou vas aan mekaar as gevolg van kleeftand op die kante van die bokse, en saam tref hulle die vloer met 'n kinetiese energie van 29 J. Bereken die aanvanklike spoed van die 3.2 kg boks.



[6]

In this problem we have two events – firstly the collision (with an initial velocity and the velocity with which the two boxes leave the table) and then secondly we have the fall to the floor. This part can be calculated with kinematics equations, but it is easier with conservation of energy.

This is a completely inelastic collision, followed by projectile motion. In the collision, we use momentum conservation.

$$\begin{aligned}\vec{p}_{1i} &= \vec{p}_{(1+2)f} \\ m_1 v_{1i} &= (m_1 + m_2) v_{edge} \\ v_{edge} &= \frac{m_1 v_{1i}}{(m_1 + m_2)}\end{aligned}$$

Using conservation of mechanical energy and taking the floor as $U_g = 0$ J

Therefore, $v = 1.8$ m/s

$$\begin{aligned}K_{edge} + U_{edge} &= K_{floor} + U_{floor} \\ \frac{1}{2} m_{(1+2)} v_{edge}^2 + m_{(1+2)} gh &= K_{floor} + 0 \\ v_{edge} &= \sqrt{\frac{2(K_{floor} - m_{(1+2)} gh)}{m_{(1+2)}}} \\ &= 1.82 \text{ m/s}\end{aligned}$$

Equating the two equations for v_{edge} :

$$\begin{aligned}v_{edge} &= \frac{m_1 v_{1i}}{(m_1 + m_2)} = \sqrt{\frac{2(K_{floor} - m_{(1+2)} gh)}{m_{(1+2)}}} \\ v_{1i} &= \sqrt{\frac{2(K_{floor} - m_{(1+2)} gh)}{m_{(1+2)}}} \frac{m_1 + m_2}{m_1} \\ &= \sqrt{\frac{2(29 \text{ J} - (5.2 \text{ kg})(9.8 \text{ m/s}^2)(0.40 \text{ m}))}{(5.2 \text{ kg})}} \frac{3.2 \text{ kg} + 2.0 \text{ kg}}{3.2 \text{ kg}} \\ &= 2.96 \text{ m/s} \\ &\approx 3.0 \text{ m/s}\end{aligned}$$

Question 8 / Vraag 8

Trucks can be run on energy stored in a rotating flywheel, with an electric motor getting the flywheel up to its top speed of 100 rev/s. Such a flywheel can be a solid uniform cylinder with a mass of 500 kg and a radius of 1.0 m rotating about an axis along the length of the cylinder and through its centre of mass.

Motortrokke kan aangedryf word met energie wat gestoor word in 'n roterende vliegwiel, wat deur 'n elektriese motor gedraai word tot 'n maksimum spoed van 100 rev/s. So vliegwiel kan 'n soliede uniforme silinder met 'n massa van 500 kg en 'n radius van 1.0 m wees, wat roteer om 'n as langs die lengte van die silinder en deur die massa-middelpunt daarvan.

- a) Determine the kinetic energy of the flywheel after charging.

Bereken die kinetiese energie van die vliegwiel nadat dit gelaai is.

[3]

Using $I = \frac{1}{2}MR^2$ for a disk then the rotational kinetic energy is

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega^2 = \frac{1}{4}(500\text{ kg})(200\pi\text{ rad/s})^2(1.0\text{ m})^2 = 4.9 \times 10^7\text{ J.}$$

-
- b) If the truck uses an average power of 8.0 kW, for how many minutes can it operate before charging is necessary?

Indien die motortrok 'n gemiddelde drywing het van 8.0 kW, vir hoeveel minute kan die trok aangedryf word voor dit herlaai moet word?

[3]

We solve $P = K/t$ (where P is the average power) for the operating time t .

$$t = \frac{K}{P} = \frac{4.9 \times 10^7\text{ J}}{8.0 \times 10^3\text{ W}} = 6.2 \times 10^3\text{ s}$$

which we rewrite as $t \approx 100\text{ min.}$ **✓ must be minutes**

Question 9 / Vraag 9

A uniform disk rotates at a constant angular velocity of 150 rev/min when an external torque acts on it. When the torque is removed, the friction of the bearings and the air stops the wheel in 2.2 h.

'n Uniforme skyf roteer teen 'n konstante hoeksnelheid van 150 rev/min wanneer 'n eksterne draaimoment toegepas word daarop. Wanneer die draaimoment verwyder word, sal die wrywing van die laers en die lug veroorsaak dat die wiel in 2.2 uur tot stilstand kom.

- a) Calculate the constant angular acceleration of the disk during the slowdown.

Bereken die konstante hoekversnelling van die skyf terwyl dit tot stilstand kom.

[3]

The angular acceleration is

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{0 - 150\text{ rev/min}}{(2.2\text{ h})(60\text{ min/1h})} = -1.14\text{ rev/min}^2.$$

OR

$$\omega = \frac{150\text{ rev}}{\text{min}} \times \frac{2\pi\text{ rad}}{\text{rev}} \times \frac{1\text{ min}}{60\text{ s}} = 15.7\text{ rad/s}$$

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{0 - 15.7\text{ rad/s}}{(2.2\text{ h})(3600\text{ s/1h})} = -1.98 \times 10^{-3}\text{ rad/s}^2.$$

- b) How many revolutions does the disk make before stopping?
Hoeveel omwentelings maak die skyf voordat dit stop?

[3]

With $t = (2.2)(60) = 132$ min, the number of revolutions is

$$\begin{aligned}\theta &= \omega_0 t + \frac{1}{2} \alpha t^2 \\ &= (160 \text{ rev/min})(132 \text{ min}) + \frac{1}{2} (-1.21 \text{ rev/min}^2)(132 \text{ min})^2 \\ &= 10.6 \times 10^3 \text{ rev.} \quad \checkmark \checkmark\end{aligned}$$

Or

$$\begin{aligned}\theta &= \omega_0 t + \frac{1}{2} \alpha t^2 \\ &= (15.7 \text{ rad/s})(7.92 \times 10^3 \text{ s}) + \frac{1}{2} (-1.98 \times 10^{-3} \text{ rad/s}^2)(7.92 \times 10^3 \text{ s})^2 \\ &= 62.2 \times 10^3 \text{ rad.}\end{aligned}$$

You can also use the formula: $\omega^2 = \omega_0^2 + 2\alpha\theta$

- c) At the instant the disk is turning at 75 rev/min, calculate the **tangential component** of the **linear** acceleration of a particle that is 50 cm from the axis of rotation.
Die oomblik wat die skyf teen 75 rev/min draai, bereken die raaklynige-komponent van die linieêre versnelling van 'n deeltjie wat 50 cm van die rotasie as geleë is.

[3]

With $r = 50 \text{ cm} = 0.500 \text{ m}$, the tangential acceleration is

$$\begin{aligned}a_t &= \alpha r \\ &= (-1.21 \text{ rev/min}^2) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) \left(\frac{1 \text{ min}}{60 \text{ s}} \right)^2 (0.500 \text{ m}) \\ &= -1.06 \times 10^{-3} \text{ m/s}^2 \quad \checkmark\end{aligned}$$

which yields $a_t = -1.06 \text{ mm/s}^2$. Note that this is independent of the given angular velocity.

ENJOY THE REST OF THE WEEKEND
GENIET DIE RES VAN DIE NAWEEK